

Complete CI/CD Setup Guide (Windows)

Automate application deployment using (Jenkins + Git + Maven + Docker)

This setup creates a CI/CD pipeline where:

- ✓ Git stores your source code (version control).
- ✓ Jenkins automatically triggers builds when code is pushed.
- ✓ Maven compiles, tests, and packages the application (e.g., JAR/WAR).
- ✓ Docker builds an image and runs the app in a container.

PHASE 1 – Install Required Software

Step 1 – Install JDK 17

1. Download and install Temurin JDK 17.

2. Set Environment Variable:

```
JAVA_HOME = C:\Program Files\Eclipse Adoptium\jdk-17
```

3. Add to PATH:

```
%JAVA_HOME%\bin
```

4. Verify:

```
java -version
```

Step 2 – Install Maven

1. Extract Maven to C:\apache-maven

2. Set MAVEN_HOME:

```
MAVEN_HOME = C:\apache-maven
```

3. Add to PATH:

```
%MAVEN_HOME%\bin
```

4. Verify:

```
mvn -v
```

Step 3 – Install Git

Install Git for Windows and verify:

```
git --version
```

Step 4 – Install Docker Desktop

Install Docker Desktop, enable WSL2, restart system.

Verify:

```
docker --version
docker ps
```

PHASE 2 – Install Jenkins

1. Download Jenkins Windows (.msi) from: <https://www.jenkins.io/download/>

2. Install with default settings.

3. Open in browser:

<http://localhost:8080>

4. Unlock using password from:

C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword

5. Install Suggested Plugins and create admin user.

PHASE 3 – Verify Tools Inside Jenkins

Create a Pipeline job named "tool-check" and paste:

```
pipeline {
    agent any
    stages {
        stage('Check Tools') {
            steps {
                bat 'java -version'
                bat 'mvn -v'
                bat 'git --version'
                bat 'docker --version'
            }
        }
    }
}
```

PHASE 4 – Create CI/CD Pipeline

Create a Pipeline job named "java-docker-ci" and paste:

```
pipeline {
```

```

agent any

stages {

    stage('Checkout Code') {
        steps {
            git branch: 'master',
              url: 'https://github.com/kodewala-
academy/30th_June_2025_Offline.git'
        }
    }

    stage('Build Application') {
        steps {
            dir('eureka-service') {
                bat 'mvn clean package -DskipTests'
            }
        }
    }

    stage('Build Docker Image') {
        steps {
            dir('eureka-service') {
                bat 'docker build -t eureka-service:1.0 .'
            }
        }
    }

    stage('Stop Old Container') {
        steps {
            bat 'docker stop eureka-container || exit 0'
            bat 'docker rm eureka-container || exit 0'
        }
    }

    stage('Run Docker Container') {
        steps {
            bat 'docker run -d -p 8761:8761 --name eureka-
container eureka-service:1.0'
        }
    }
}

```

Required Dockerfile (Inside eureka-service folder)

```

FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY target/*.jar app.jar

```

EXPOSE 8761

ENTRYPOINT ["java","-jar","app.jar"]

Common Errors & Fixes

1. Repository not found – Use correct .git URL.
2. Dockerfile not found – Ensure Dockerfile exists in eureka-service folder.
3. openjdk:17 not found – Use eclipse-temurin:17-jdk image.
4. Port already allocated – Stop old container using docker stop and docker rm.